Executive summary: momentum with a 45% shallower drawdown than 60/40

Hanyu Zhang · Columbia University · October 5, 2026

−17.91%
worst daily drawdown
60/40 −32.62%; SPY −55.19%

−5.0%
return in 2022
60/40 −16.7%; SPY −18.2%

0.38
CAGR ÷ max drawdown
60/40 0.25; SPY 0.20

6.89%
annual return (CAGR)
Sharpe 0.614; vol 9.26%

Strategy

Each month, hold the three strongest of eight liquid ETFs, keep each only if it beats cash, and scale exposure to a 10% volatility cap. Long-only; no leverage.

Investor benefit

Participation in trends across equities, bonds, gold and commodities, with materially smaller losses in crises. Fully rules-based and auditable month by month.

The ask

A **$1m** launch mandate to validate live execution, scaling to **$10m** once independent data and execution checks are complete.

Backtest, 2007-08-31 to 2026-08-31 (first invested return 2007-09-04), net of 10 bps per traded risky dollar including initial purchases. Sharpe from monthly excess returns. Source: `outputs/tables/performance.csv`, `stress_periods.csv`.

Momentum landscape: an established effect, few options for long-only allocators

Approach	Signal and positions	Risk control	Limitation for allocators	Our position
Equity long/short (cross-sectional)	Rank stocks on past return; long winners, short losers	Optional dynamic weighting	Crashes when losers rebound after panics [DM16]	Research input (A)
Futures trend following	Sign of each contract's own return; long or short	Inverse-volatility sizing	Requires futures, leverage and shorting [BK13]	Not required
Relative ETF momentum	Rank assets against peers; long top k	Diversification only	Fully invested through broad sell-offs	Component (B1)
Dual momentum with volatility cap	Relative rank *and* own return versus cash	Cash filter plus portfolio volatility cap	Gives up some upside	**Proposal (B3)**

- Momentum is documented across stocks, bonds, currencies and commodities [AFIM14]. Relative momentum chooses *which* assets to hold; absolute momentum decides *whether* to be invested [KW15; Lecture].
- **Our niche:** the trend evidence delivered through eight liquid US-listed ETFs, long-only, unlevered, with an explicit risk cap that an allocator can monitor.

Bracketed tags refer to the references in Appendix 7.

Research foundation: momentum risk can be managed, and B3 makes it investable

A. Factor research (1990–2026)

- Applies bear-market, VIX and volatility controls to the US stock momentum factor.
- Static factor max drawdown −63.29%; combined controls (A3) −18.89%.
- Lesson: momentum's worst losses cluster in high-volatility states, so scaling exposure by risk can sharply reduce drawdowns [DM16].
- Synthetic and gross of stock-level turnover, financing and borrow costs. Shown in the appendix as research evidence, not as an offer.

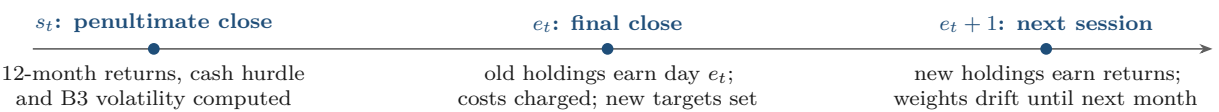
B. Investable ETF strategy (2007–2026)

- Explicit holdings in eight ETFs, priced from adjusted closes, with trading costs charged.
- Built in steps: equal weight (B0), relative momentum (B1), cash hurdle (B2), volatility cap (B3).
- Applies the same risk-scaling principle in a long-only, unlevered form.
- B3 is the strategy we propose.

A and B differ in sample, exposure and cost treatment, so their returns are not compared. B is a separate cross-asset application of momentum principles, not an implementation of A's equity long/short portfolio.

Data and timing: signals at the penultimate close, trades at the final close

Series	Provider	Coverage used	Treatment
8 ETF prices	Yahoo Finance (yfinance 1.7.0)	Common history from 2006-02-06 (DBC binds)	Adjusted close as total return; dividends not double counted
Cash	Kenneth French daily RF	Through 2026-08-31 (binding end date)	Compounded daily
Factor (A)	French daily Mom	1926-11-03 to 2026-08-31	Kept separate from monthly Mom
VIX (A)	Cboe official closes	From 1990-01-02	Matched to prior available close



- 12-month signals plus a 126-day volatility history give a common first execution on 2007-08-31 for every ETF strategy and benchmark.
- One full session separates the signal from the trade, so no decision uses same-day information.
- Current data vintages may revise history, and the ETF universe was chosen with hindsight; both are standard backtest caveats.

Sources: `outputs/tables/data_coverage.csv`; `data/manifests/inputs.json`; `research/data_quality.md`.

Portfolio construction: B0, B1 and B2

Universe $\mathcal{U} = \{\text{SPY, EFA, EEM, IEF, TLT, GLD, DBC, VNQ}\}$; P_i = adjusted close; s_t = penultimate trading day of month t.

Signals at s_t

$$M_{i,t} = \frac{P_i(s_t)}{P_i(s_{t-12})} - 1,$$
$$C_t = \prod_{d \in (s_{t-12},\, s_t]} (1 + \mathrm{RF}_d) - 1.$$

C_t spans exactly the signal window. Rank $R_{i,t}$ by descending $M_{i,t}$; ties alphabetical.

Targets executed at e_t

$$w_{i,t}^{B0} = \tfrac{1}{8},$$
$$w_{i,t}^{B1} = \tfrac{1}{3}\,\mathbf{1}[R_{i,t} \le 3],$$
$$w_{i,t}^{B2} = \tfrac{1}{3}\,\mathbf{1}[R_{i,t} \le 3]\,\mathbf{1}[M_{i,t} > C_t],$$
$$w_t^{\text{cash}} = 1 - \textstyle\sum_i w_{i,t}.$$

- **B0**: equal weight, reset monthly.
- **B1**: three fixed one-third slots for the top three ETFs.
- **B2**: a slot that fails the cash hurdle stays in cash. No refill from rank 4 and no renormalization, so the filter genuinely reduces risk.
- Monthly snapshots; the latest month is included.
- Benchmarks: 60/40 = 60% SPY, 40% IEF reset monthly; SPY bought and held.

Every decision is logged in `outputs/audits/etf_signals.csv` (lookback dates, $M_{i,t}$, C_t, targets).

B3 risk control: exposure scales down automatically when volatility rises

Let x_d = daily gross B2 return minus RF, from B2's realized history. At s_t:

$$\hat{\sigma}_t = \sqrt{252}\ \mathrm{sd}\left(x_{d-125}, \ldots, x_d\right), \quad d = s_t,$$

$$q_t = \min\left(1,\ \frac{0.10}{\max(\hat{\sigma}_t,\ 0.05)}\right),$$

$$w_{i,t}^{B3} = q_t\, w_{i,t}^{B2}, \qquad w^{\mathrm{cash}} = 1 - q_t \textstyle\sum_i w_{i,t}^{B2}.$$

- 126 daily observations, sample standard deviation.
- $q_t \leq 1$: the cap only reduces risk and never adds leverage.
- On average B3 was 77% invested, holding 22.8% in cash.

Worked example (illustrative)

Top-three signals 12%, 8%, 1%; matched cash 2% ⇒ two slots pass. With $\hat{\sigma}_t = 16\%$, $q_t = 0.10/0.16 = 0.625$.

Slot	B1	B2	B3
Rank 1 (12%)	33.33%	33.33%	20.83%
Rank 2 (8%)	33.33%	33.33%	20.83%
Rank 3 (1%)	33.33%	0.00%	0.00%
Cash	0.00%	33.33%	58.33%

B3 cash = failed slot (33.33%) + scaling cut $(1 - 0.625) \times 66.67\% = 25.00\%$.

Historical q_t and $\hat{\sigma}_t$ per rebalance: `outputs/audits/etf_risk_sizing.csv`.

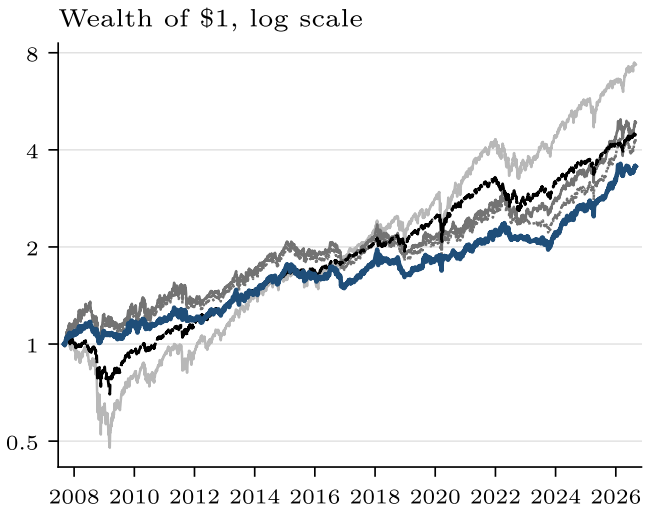
Results: B3 has the shallowest drawdown and the best return per unit of drawdown

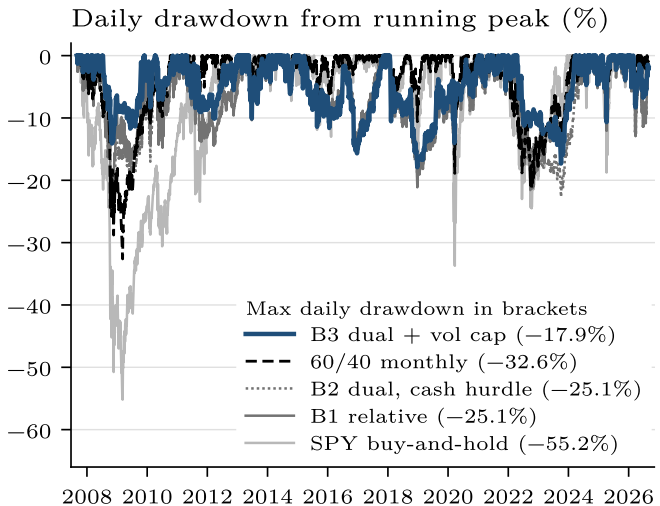
Code	Strategy	CAGR	Vol	Sharpe	Max daily DD	Worst month	CAGR/MDD
B0	Equal weight	6.40%	11.09%	0.487	−37.92%	−17.3%	0.17
B1	Relative momentum	8.66%	12.73%	0.607	−25.12%	−14.4%	0.34
B2	Dual momentum, cash hurdle	7.95%	12.29%	0.570	−25.15%	−14.4%	0.32
B3	**Dual momentum + vol cap**	6.89%	9.26%	0.614	**−17.91%**	**−8.2%**	**0.38**
60/40	Monthly 60/40	8.16%	9.56%	0.719	−32.62%	−10.3%	0.25
SPY	SPY buy-and-hold	11.08%	15.58%	0.662	−55.19%	−16.5%	0.20

- **Each step adds value.** Relative selection lifts CAGR from 6.40% to 8.66% and cuts the worst drawdown from −37.92% to −25.12%. The volatility cap then takes it to −17.91%, the clearest single improvement.
- **Versus 60/40:** 45% shallower worst drawdown and a smaller worst month, for 1.3 points less annual return. 60/40 keeps a higher Sharpe; B3 is built for investors who rank drawdown control first.
- The cash hurdle (B2) contributes little to the worst drawdown in this sample; the cap does the heavy lifting.

Identical dates for all rows: 2007-08-31 to 2026-08-31, net of 10 bps per traded risky dollar. CAGR/MDD = CAGR ÷ |max daily drawdown|. Source: `outputs/tables/performance.csv`.

B3 compounds steadily while avoiding the deep troughs of 60/40 and equities





2007-08-31 (first return 2007-09-04) to 2026-08-31; net of 10 bps per traded risky dollar; drawdown peaks include initial capital. Daily NAV from `outputs/tables/*_daily.csv`. B0 not shown (max DD −37.92%).

Consistency: B3 limited losses in both crisis periods and led over 2020–2026

	B3 vol-capped dual momentum			Monthly 60/40		
Period	CAGR	Sharpe	Max daily DD	CAGR	Sharpe	Max daily DD
Development 2007-09 to 2013-12	6.2%	0.66	−14.1%	6.4%	0.61	−32.6%
Validation 2014–2019	3.8%	0.35	−17.9%	8.8%	1.22	−10.6%
Historical holdout 2020-01 to 2026-08	10.5%	0.81	−17.3%	9.3%	0.59	−21.3%
Full sample 2007-09 to 2026-08	6.9%	0.61	−17.9%	8.2%	0.72	−32.6%

Stress measure	B3	60/40	SPY
Max daily DD 2007–13 (incl. 2008–09)	**−14.1%**	−32.6%	−55.2%
Worst three months	**−11.1%**	−16.3%	−29.6%
Return in 2020	**7.8%**	16.0%	18.3%
Return in 2022 (stocks and bonds fell)	**−5.0%**	−16.7%	−18.2%

- Smaller drawdowns than 60/40 in 2007–2013 and 2020–2026. In calm 2014–2019 markets, 60/40 did better.
- 2020–2026: higher return *and* Sharpe than 60/40. These years were reviewed during research, so we treat them as supporting, not out-of-sample, evidence.
- B3 lags in strong rallies (2020); that is the price of protection.

Paired block bootstrap (6-month blocks, 1,000 draws): B3−60/40 Sharpe difference −0.106, 95% interval [−0.598, 0.279]; the case rests on drawdown, not Sharpe. Sources: `performance.csv`, `stress_periods.csv`, `bootstrap.csv`.

Robustness: drawdown control holds across costs, timing and breadth

B3 variant (one change)	CAGR	Δ pp	Sharpe	Max DD
Frozen rule, sensitivity dates	6.53%	–	0.579	−17.91%
Exclude GLD	5.28%	−1.25	0.456	−20.16%
Skip latest month (12–1)	6.46%	−0.07	0.562	−24.73%
Costs 25 bps per traded $	6.03%	−0.50	0.526	−18.24%
Costs 50 bps per traded $	5.20%	−1.33	0.438	−18.92%
Execute one session later	6.99%	+0.46	0.624	−17.36%
Top 2 slots	6.44%	−0.09	0.545	−19.71%
Top 4 slots	6.39%	−0.14	0.594	−15.45%
6-month lookback (diagnostic)	7.39%	+0.86	0.691	−21.60%
9-month lookback (not adopted)	8.24%	+1.72	0.833	−14.27%

- Max drawdown stays between −14.3% and −24.7% in every variant.
- Trading one session later changes almost nothing; even 50 bps costs leave CAGR at 5.2%.
- Known dependencies: gold (GLD) diversification and including the latest month.
- The 9-month signal tested better, but we keep the pre-specified 12-month rule to avoid fitting to history.

Common sensitivity sample 2007-10-31 to 2026-08-31 (needed for the skip-month warm-up), so the first row differs from the headline. Source: `outputs/tables/sensitivities.csv`; all 84 evaluated rows in `variant_ledger.csv`.

Implementation: liquid ETFs, low costs and a staged mandate

Costs

- One rebalance a month; turnover 3.15× a year (buys + sells).
- At 10 bps per traded dollar, about 0.31% a year; gross CAGR 7.23% vs net 6.89%.
- ETF expense ratios are already inside the returns. Even at 50 bps, B3 keeps its drawdown control (slide 10).

Mandate path

1. **$1m launch**: every historical trade below 2% of average daily volume.
2. **$10m**: after independent data checks and measured closing-auction costs, with DBC orders sized or split as needed.

Trade size relative to ADV20, B3

	$1m	$10m scenario			
ETF	Max	Max	p95	Trades	>5%
DBC	1.92%	19.17%	3.29%	67	13.4%
IEF	0.77%	7.72%	2.25%	50	8.0%
VNQ	0.31%	3.11%	1.92%	114	0.0%
TLT	0.10%	1.05%	0.53%	80	0.0%
EFA	0.10%	0.98%	0.43%	84	0.0%

SPY, EEM, GLD and EFA never exceed 1% of ADV20 even at $10m. The constraint is DBC, which peaks at **19.17%** at $10m: the reason the mandate is staged.

Participation = AUM × NAV × $|\Delta w|$ / 20-day average dollar volume to the prior signal close; all-day volume, not closing-auction volume. Sources: `capacity_summary.csv`, `capacity_active_trades_summary.csv`.

Investment proposal: who it is for, the risks, and next steps

Why allocate to B3

- **Drawdown control**: −17.91% worst drawdown vs −32.62% for 60/40 and −25.12% for relative momentum.
- **Diversifies stock–bond risk**: −5.0% in 2022 vs −16.7% for 60/40.
- **Transparent**: published rules, liquid ETFs, every signal, trade and NAV auditable; no leverage, no shorting.

Suited to

Allocators who value smaller losses over maximum return and can accept periods of lagging in strong rallies.

Key risks and mitigants

Risk	Mitigant
Fast reversals, whipsaw	Monthly rebalancing; volatility cap
Lags in sharp rallies	Disclosed: 2020 7.8% vs 60/40 16.0%
Backtest overfitting	Rules frozen in advance; 9-month variant not adopted
Capacity in DBC	Staged mandate from $1m

Next steps

1. Independent reconciliation of prices and distributions (e.g., CRSP).
2. Measure closing-auction costs; begin time-stamped paper trading.
3. Launch the $1m mandate; review for scale-up to $10m.

Backtested, net of modeled costs; past performance does not guarantee future results. Sources: `performance.csv`, `stress_periods.csv`.

Appendix contents

A: factor research (synthetic gross)

1. A0–A3 construction and conventions
2. A results and diagnostics
3. A wealth and drawdown

Methods and precision

4. Trading, cost and metric conventions
5. B3 exposure management
6. Five-decimal results table

Sources

7. References
8. Data provenance and reproducibility

A results are synthetic, gross and dated 1990-01-31 to 2026-08-31. They are shown for research context and are not comparable with the ETF results in the main deck.

Appendix 1. Factor overlay: A0–A3 construction

Daily synthetic capital return, with m_d = daily French Mom (already an excess spread) and f_d = RF:

$$r_d = f_d + a_t\, m_d - \text{cost}_d.$$

At the penultimate close s_t:

$$\begin{aligned}
\sigma_t^{\text{Mom}} &= \sqrt{252}\ \text{sd}(\text{last } 126\ m_d),\\
a_t^{\text{vol}} &= \min\Big(1.5,\ \frac{0.10}{\max(\sigma_t^{\text{Mom}}, 0.05)}\Big),\\
\text{Bear}_t &= \mathbf{1}\big[\text{TR}^{\text{mkt}}(s_t)/\text{TR}^{\text{mkt}}(s_{t-24}) < 1\big],\\
\text{HighVol}_t &= \mathbf{1}[\text{VIX}(s_t) > 25],\\
g_t &= 1,\ 0.5,\ 0 \text{ for } 0,\ 1,\ 2 \text{ active flags.}
\end{aligned}$$

Code	Exposure a_t	Role
A0	1	Static factor
A1	a_t^{vol}	Volatility scaling
A2	g_t	Bear/VIX regime
A3	$a_t^{\text{vol}}\, g_t$	Combined

- Exposure held until the next monthly execution. RF is not subtracted twice.
- One factor unit ≈ \$1 long + \$1 short per \$ of capital; the 1.5 cap allows gross notional up to 3×.
- The bear/VIX rule is a heuristic motivated by panic-state evidence [DM16, pp. 226–227; KW15, pp. 59–60]; it is not D&M's conditional mean–variance strategy.
- Gross. Illustrative charges ($2|\Delta a|\times$cost, annual drags) omit trading inside Mom and borrow.

Execution 1990-01-31, first return 1990-02-01, end 2026-08-31. Monthly exposure decisions in `outputs/audits/factor_decisions.csv`.

Appendix 2. Factor results (synthetic, gross of stock-level costs)

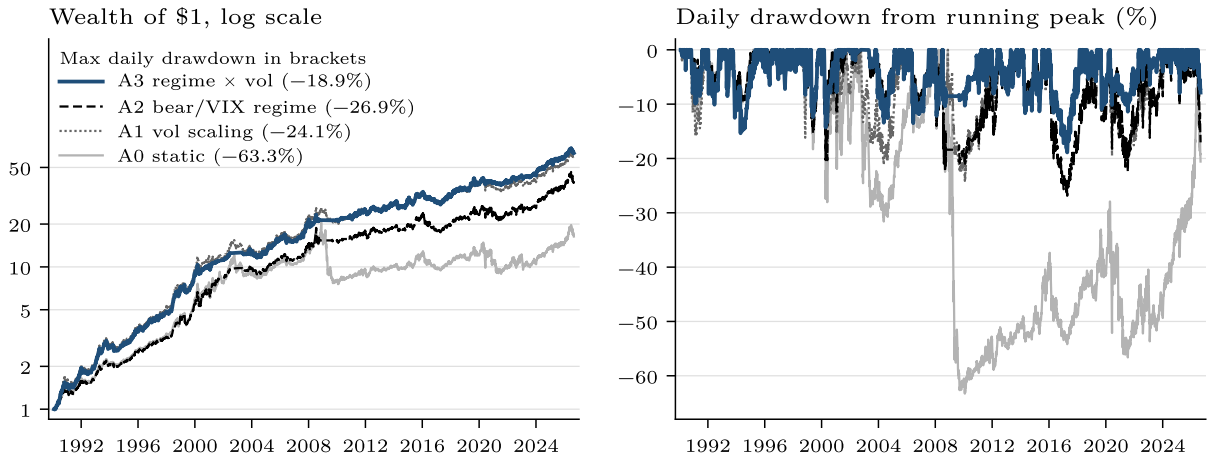
Code	Strategy	CAGR	Ann. vol	Sharpe	Max daily DD	Avg exposure	Exposure turnover/yr
A0	Static factor	7.94%	16.17%	0.393	−63.29%	1.00	0.05
A1	Volatility scaling	11.76%	12.44%	0.751	−24.13%	0.99	1.28
A2	Bear/VIX regime	10.50%	11.48%	0.702	−26.87%	0.84	2.08
A3	Regime × volatility	11.99%	10.45%	0.894	−18.89%	0.89	2.35

	A0 static			A3 combined		
Period	CAGR	Sharpe	DD	CAGR	Sharpe	DD
Development	10.1%	0.48	−63.3%	14.9%	1.04	−15.4%
Validation	3.1%	0.24	−26.9%	5.5%	0.47	−18.9%
Holdout	4.7%	0.19	−39.8%	7.9%	0.74	−11.4%

- Both flags active at 6.8% of monthly decisions; any flag in 25%; the 1.5 cap binds in 22.5%.
- March–May 2009: A0 lost 45.6%; A2/A3 had zero factor exposure. That is de-risking, not a hedge.
- A3−A0 Sharpe difference 0.501, bootstrap interval [0.208, 0.760], before any missing costs.

Basis: cash + factor exposure, gross; underlying stock turnover, financing and borrow costs not measured. Sources: `performance.csv` (prototype A), `factor_sizing_diagnostics.csv`, `stress_periods.csv`, `bootstrap.csv`.

Appendix 3. Factor wealth and drawdown (synthetic gross, 1990–2026)



Daily files `outputs/tables/A{0,1,2,3}_daily.csv`. Execution 1990-01-31 (first return 1990-02-01) to 2026-08-31. Synthetic cash + daily French Mom exposure, **gross**. Peaks include initial $1. Historical crash protection motivates research; it does not establish implementability.

Appendix 4. Trading, cost and metric conventions

Daily drift and rebalancing

$$
\begin{aligned}
R_d^{\text{gross}} &= \textstyle\sum_i w_{i,d-1}\, r_{i,d} + w_{d-1}^{\text{cash}}\mathrm{RF}_d,\\
\tilde{w}_{i,d} &= w_{i,d-1}(1 + r_{i,d})/(1 + R_d^{\text{gross}}),\\
T_t &= \textstyle\sum_{i \neq \text{cash}} \left| w_{i,t}^{*} - \tilde{w}_{i,e_t} \right|,\\
V^{\text{after}} &= V^{\text{before}}(1 + R_{e_t}^{\text{gross}})(1 - c\,T_t), \quad c = 0.001.
\end{aligned}
$$

Old holdings earn the execution-day return, then the cost is charged and post-cost NAV is allocated to targets. A full switch has $T = 2$ (20 bps). Initial purchases are charged.

Example: 50/50 in X/Y, X +10% ⇒ weights 52.38/47.62%; moving to 100% Y gives $T = 1.0476$, cost 10.48 bps.

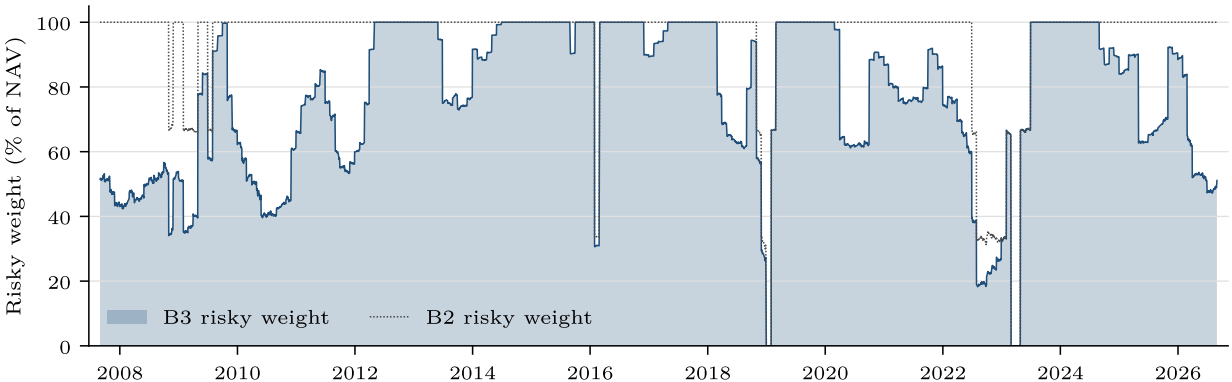
Metrics

$$
\begin{aligned}
\text{CAGR} &= V_{\text{end}}^{365.25/\text{days}} - 1 \quad \text{(from execution date)},\\
\text{Sharpe} &= \sqrt{12}\,\overline{x}/\,\mathrm{sd}(x),\\
\text{MDD} &= \min_t \left(V_t / \max_{s \le t} V_s - 1 \right),\ V_0 = 1.
\end{aligned}
$$

- x = compounded monthly capital return minus compounded daily RF; sample s.d.
- Annual vol = $\sqrt{12}$ sd of monthly returns.
- The initial cost stays in daily NAV and enters the first full month for monthly statistics.
- Daily and month-end drawdowns are separate fields; the deck reports daily.
- Adj Close includes distributions; nothing added twice.

Independent reconciliation of trades, weights and NAV: `outputs/audits/independent_accounting_checks.csv`.

Appendix 5. B3 exposure management over time



- Average cash: B3 22.8% vs B2 5.2%. The cap bound most persistently in 2007–2011; the deepest cash periods (late 2018–2019, 2022–23) came mainly from slots failing the cash hurdle.
- Exposure falls in turbulent markets and returns toward 100% in steady trends. Lower exposure in high-volatility periods is the main source of B3's smaller drawdowns.

End-of-day weights from `outputs/tables/B3_daily.csv` and `B2_daily.csv` (risky weight = 1 − cash weight). Execution 2007-08-31 to 2026-08-31.

Appendix 6. Five-decimal results (decimals, not percentages)

Code	Start	Basis	CAGR	Ann. vol	Sharpe	Sortino	Max daily DD	Turnover/yr
B0	2007-08-31	net 10 bps	0.06405	0.11092	0.48708	0.70546	−0.37924	0.39602
B1	2007-08-31	net 10 bps	0.08658	0.12729	0.60665	0.92946	−0.25123	3.58982
B2	2007-08-31	net 10 bps	0.07953	0.12294	0.57005	0.86930	−0.25147	3.53961
B3	2007-08-31	net 10 bps	0.06891	0.09264	0.61357	0.97000	−0.17911	3.14944
60/40	2007-08-31	net 10 bps	0.08165	0.09559	0.71919	1.08247	−0.32617	0.27154
SPY	2007-08-31	net 10 bps	0.11077	0.15583	0.66228	0.99297	−0.55189	0.05262
Cash	2007-08-31	net 10 bps	0.01451	0.00569	–	–	0.00000	0.00000
A0	1990-01-31	synthetic gross	0.07938	0.16174	0.39282	0.56166	−0.63288	0.05467
A1	1990-01-31	synthetic gross	0.11762	0.12442	0.75071	1.28154	−0.24129	1.27534
A2	1990-01-31	synthetic gross	0.10500	0.11476	0.70178	1.21233	−0.26873	2.07746
A3	1990-01-31	synthetic gross	0.11994	0.10450	0.89354	1.66255	−0.18893	2.34986

Values are decimals: 0.06891 means 6.891%. All B rows end 2026-08-31 with the same execution start; A rows are synthetic gross with a different start and are not comparable with B. Cash has no Sharpe or Sortino because its excess return is zero by construction. For A, turnover is change in factor exposure, not stock turnover.

Source: `outputs/tables/performance.csv`, sample `full`; rounded from full precision by `scripts/build_assets.py`.

Appendix 7. References

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Kolanovic, M., & Wei, Z. (April 2015). *Momentum Strategies Across Asset Classes: Risk Factor Approach to Trend Following.* J.P. Morgan. pp. 19–20, 59–60. [KW15]

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Data providers: Kenneth R. French Data Library (daily and monthly Mom, Fama–French factors, RF); Cboe VIX historical closes; Yahoo Finance via yfinance 1.7.0. Methodology URLs and access dates are in `research/source_map.md`.

Page and slide locations follow `research/source_map.md` of the original package. Readings support motivation and method; they do not supply this portfolio's results.

Appendix 8. Data provenance and reproducibility

Original research package (unchanged)

- `data/raw`, `data/manifests`: snapshots, URLs, timestamps, SHA-256.
- `config/base.yaml`, `research/freeze.json`: frozen rules.
- `src/`: shared backtest code; `notebooks/`: executed walk-throughs.
- `outputs/tables`: canonical returns and metrics; `outputs/audits`: signals, targets, trades, holdings, clean-run hashes.

Reproduce (Python 3.12, cached data, offline): package command-line interface, `run --offline` `python scripts/clean_reproduce.py`

Presentation build

- `scripts/build_assets.py` reads only `outputs/tables/*.csv`, writes `generated/*.tex` and `figures/*.pdf`, and asserts the headline values and chart drawdowns against `performance.csv`.
- Build: `python3 scripts/build_assets.py`, then `pdflatex` twice.

Data caveats: public caches for personal academic use; current vintages may be revised; universe chosen retrospectively.